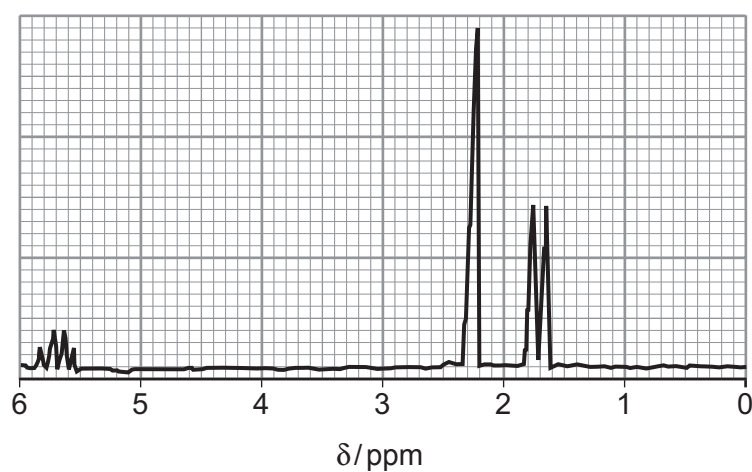
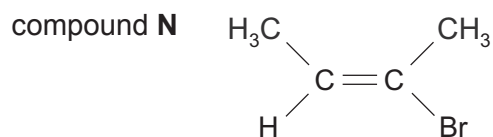
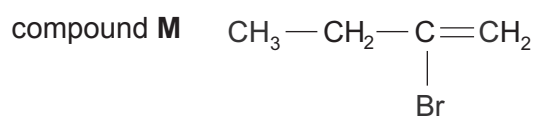
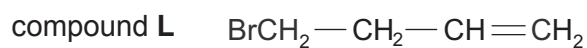


13. (a) The analysis of a compound shows that its formula is C_4H_7Br .
The high resolution 1H NMR spectrum of the compound is shown below.



- (i) A student suggested that the compound might be one of the following.



Discuss the splitting pattern seen in the 1H NMR spectrum to decide whether the compound is **L**, **M** or **N**. Give reasons for your answer. [4]

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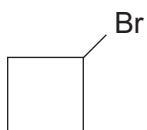
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- (ii) Another student suggested that the compound was bromocyclobutane.



- I. Describe a chemical test that would show that the compound was either **L**, **M** or **N** and not bromocyclobutane. You should state the result of your test with each compound. [1]

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- II. Explain how the ^{13}C NMR spectrum would show that the compound was **L**, **M** or **N** and not bromocyclobutane. [2]

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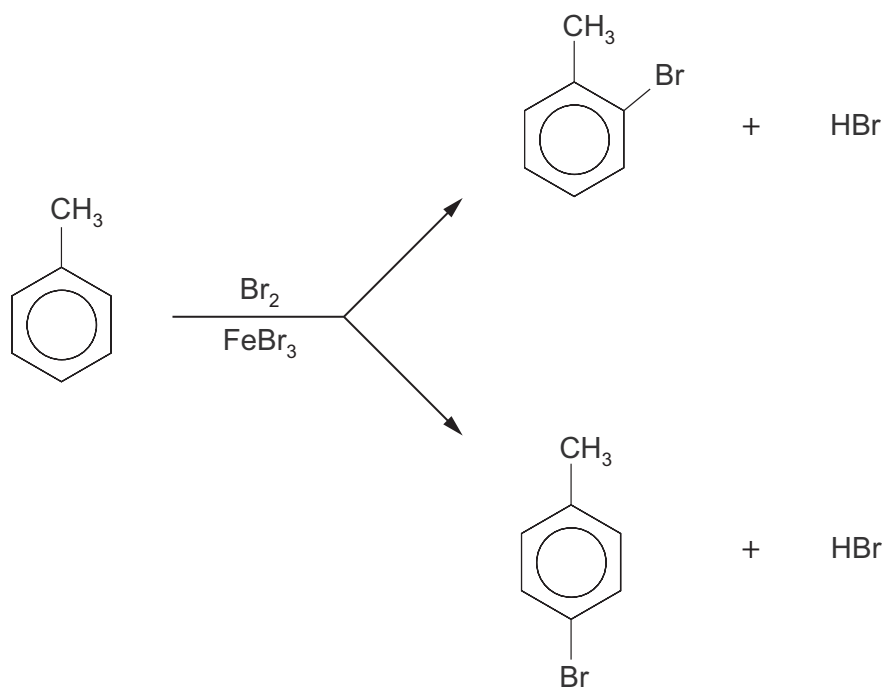
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- (b) The bromination of methylbenzene using electrophilic substitution gives a mixture of 2-bromomethylbenzene and 4-bromomethylbenzene.

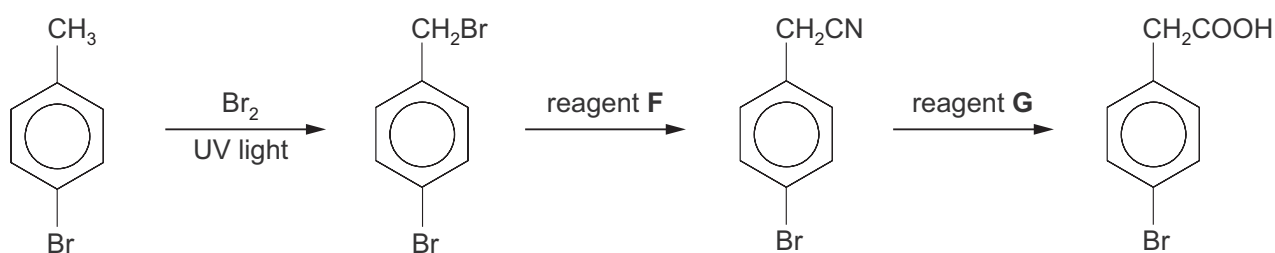


Give the mechanism of the reaction that produces 4-bromomethylbenzene.

[3]



(c) A reaction sequence to produce 4-bromophenylethanoic acid is shown below.



- (i) The bromination of 4-bromomethylbenzene is a free radical process.

Give the equation for this reaction and then use the equation to calculate the minimum mass of bromine needed to convert 0.15 mol of 4-bromomethylbenzene to 4-bromo-1-(bromomethyl)benzene in the first stage of this reaction sequence.

[2]

Mass of bromine = g

- (ii) State the name of reagent **F**. [1]

.....

- (iii) State the name of reagent **G**. [1]

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END OF PAPER

